

# Output-sensitive sparse factorization over finite fields from dense factorization

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## Abstract

We give a deterministic reduction for complete factorization of a sparse univariate polynomial over an explicitly represented finite field. Its cost is polynomial in the term count, binary exponent length, field encoding, and total degree of the distinct densely returned factors, including arbitrary binary multiplicities. The reduction maintains a sparse numerator and a dense denominator during characteristic- $p$  descent. The numerator degree shrinks by  $p$ , while the denominator degree grows by at most the original radical degree per step. Classical logarithmic derivatives, Padé reconstruction, residue multiplicity arguments and Cartier sections supply the individual tools. We also detail the established dense factorization and irreducible construction reductions of Berlekamp and Shoup. Substitution of OpenAI’s prime-field theorem gives uniform deterministic polynomial bit complexity, without GRH, randomness or an integer-factorization oracle. That input requires the companion general cyclotomic Hecke zero-free theorem; this paper neither reproves its breakthrough nor claims formal verification.

## 1 Problem, provenance and exact model

Represent  $p$  in binary, put  $L = \lceil \log_2 p \rceil$ , and supply  $h \in \mathbf{F}_p[T]$  monic of degree  $m \geq 1$ . Primality of  $p$  and irreducibility of  $h$  are promises, or may be checked deterministically. The field  $K = \mathbf{F}_p[T]/(h)$  has power basis  $1, t, \dots, t^{m-1}$ ; every coordinate is a canonical residue with  $L$  bits. A nonzero polynomial  $f \in K[X]$  of degree  $n$  is a dense list. Complete factorization returns its leading coefficient  $c$  and distinct monic irreducibles  $g_i$  with positive binary multiplicities  $e_i$ , such that  $f = c \prod_i g_i^{e_i}$ . Constants have an empty list. Zero has no such finite unique factorization and is rejected. The field and coefficient input uses  $O((n+2)mL)$  bits; output coefficients including the leading coefficient use  $O((n+1)mL)$  and multiplicities  $O(n \log(n+1))$  bits.

For construction, a positive degree  $d$  may be supplied in binary, but the requested dense output has  $(d+1)m$  coefficient slots. Polynomial in  $d$  is polynomial in this output size, and is not polynomial in  $\log d$ . No practical efficiency is inferred from a fixed polynomial exponent. A dense-input convention cannot be inferred from a field cardinality alone; the supplied defining polynomial is part of the input.

Berlekamp’s reduction from general finite fields to split prime-field polynomials is already public [9]. Shoup’s construction reduction is public in Theorems 3.1 and 4.1 of [10]. The particular degree- $md$  lift below also occurs in Algorithm 2 and Theorem 4.1 of Rai [11]. The newly supplied input is OpenAI’s prime-field theorem [12]. We attribute that breakthrough to its source. Consolidating its implications does not independently solve the base deterministic factorization problem or invent these reductions.

Our additional problem has sparse input  $f = \sum_{i=1}^t a_i X^{n_i}$ , with distinct nonnegative binary exponents, nonzero coefficients,  $N = \max n_i$  and  $b = \max(1, \lceil \log_2(N+1) \rceil)$ . Its input size is  $O((m+1)L + t(mL+b))$ . Write  $D = \sum_i \deg g_i$  for the unweighted degree of its distinct output factors. Dense factor coefficients use  $O((D+1)mL)$  bits, and binary multiplicities  $O(Db)$  bits. The output degree  $D$ , rather than the possibly exponential  $N$ , governs the following reduction. It is not a polynomial bound in sparse output term count, nor an algorithm for finding only a small common divisor.

Padé numerator/denominator reconstruction is classical [1, Section 8]; partial radicals from logarithmic derivatives occur explicitly in [2, Section 6.3]. Our known-factor multiplicity procedure is a constructive form of Mattarei's residue argument [3, Theorem 2], with related repeated-root coding results [4]. The Cartier pull-out identity is also established [5, Proposition 1.9]. The point of the new reduction is the repeated sparse/dense representation and its degree bound, which handles all multiplicities. The inspected sparse factoring algorithms depend on numeric total input degree [6]; large-characteristic lacunary power algorithms [7] and characteristic- $p$  partial-radical bounds [2] have different scopes. We make no assertion of being first. The characteristic- $p$  residue descent itself is established as well: Lecerf's algorithms use dense coefficient vectors and numeric input-degree bounds [8, Sections 2.1–2.2]. We instead implement that descent through a compressed rational representation with the degree invariant below.

## 2 Complete sparse factorization

Let DF denote complete dense factorization over the represented  $K$ . This section is a reduction theorem independent of how that oracle is implemented.

**Theorem 1** (Sparse reduction). *For every nonzero sparse input in the above model, complete factorization reduces deterministically to at most  $2b$  calls to DF, each of degree at most  $(b+1)D$ , with polynomial nonoracle bit cost in  $t, b, D, m, L$ . No condition on the multiplicities is required. With the direct prime-Frobenius reduction below, this uses at most  $2b(b+1)D$  prime-field calls, each degree at most  $(b+1)D$ . Constants and monomials are handled without these calls.*

### 2.1 Exact multiplicity of a known factor

Let  $g \neq X$  be a known monic irreducible of degree  $\delta$ . Its formal root  $\alpha$  in  $E = K[Z]/(g)$  is nonzero. Separability of  $g$  identifies its multiplicity with the order at  $Y = 0$  of

$$f(\alpha(1+Y)) = \sum_i w_i(1+Y)^{n_i}, \quad w_i = a_i \alpha^{n_i}.$$

Keep the integer exponents; reducing them modulo the order of  $\alpha$  would destroy multiplicity information. At each node group  $n = pq + r$ , giving  $F(Y) = \sum_r (1+Y)^r H_r(Y^p)$ . Recursively compute each child's initial order  $\nu_r$  and nonzero leading coefficient  $c_r$ . Put  $v = \min \nu_r$  and

$$P(Y) = \sum_{\nu_r=v} c_r(1+Y)^r.$$

If  $s$  residues are active,  $s \leq p$  and the matrix  $[\binom{r}{j}]_{0 \leq j < s}$  is invertible: its determinant is the Vandermonde in the distinct residues divided by  $\prod_{j < s} j!$ . Thus the first nonzero coefficient  $P_j$  has  $j < s \leq p$ . Inactive terms start at order at least  $p(v+1)$ , so the parent initial term is  $(pv+j, P_j)$ . A single monomial has order zero. Induction proves the procedure and gives the useful bound

$$\text{multiplicity of } g \text{ in } f \pmod{p} \leq t-1. \tag{1}$$

This is inherited residue structure [3], not a new multiplicity/sparsity theorem. For  $g = X$ , the answer is  $\min n_i$ .

Use an explicit postorder stack. There are at most  $b$  exponent-digit levels and  $t(b+1)$  occupied nodes; the sum of squared support sizes is at most  $t^2(b+1)$ . Compute binomial columns by their recurrence, using only denominators  $j < p$ , and inspect at most  $s$  coefficients, never all  $p$  residues. Initial powering takes  $O(tb)$  multiplications in  $E$ ; remaining arithmetic takes  $O(t^2(b+1))$  additions and prime-scalar multiplications there. Since  $E$  has  $\delta$  coordinates over  $K$ , a conservative bound is  $O((t+t^2)(b+1)\delta^2)$  operations in  $K$ , plus polynomial integer work. Its prime-coordinate dimension is  $m\delta$ .

## 2.2 Recovering visible factors by an exact identity

Assume  $U(0) \neq 0$ . Write the reduced rational function  $U'/U = P/G$ , where  $G$  is monic and  $\deg P < \deg G$ . Factoring formally over  $K$  shows that  $G$  is exactly the squarefree product of factors whose multiplicities are not divisible by  $p$ . In particular  $G(0) \neq 0$ .

For  $B = 1, 2, 4, \dots$ , compute the first  $2B$  Taylor coefficients of  $U'/U$  at zero. Formal series division uses only coefficients of  $U$  through index  $2B$ , read directly from the sparse list. Solve the  $B$  constraints  $(Q \cdot \text{series})_j = 0$ ,  $B \leq j < 2B$ , with  $Q_0 = 1$  and  $\deg Q \leq B$ . The augmented matrix has  $B$  rows and  $B+1$  columns over  $K$ . Let  $A$  be the first  $B$  coefficients of the product. Accept only after checking the exact sparse identity

$$U'Q = UA. \quad (2)$$

Each side has at most  $t(B+1)$  terms before combining equal exponents. Cancel  $\gcd(A, Q)$  and make the denominator monic.

For  $B \geq \deg G$ , a solution exists. Every solution represents  $P/G$ :  $AG - PQ$  vanishes to order  $2B$  and has degree less than  $2B$ . Thus acceptance occurs by  $B \leq 2\max(1, \deg G)$ ; a vanishing prefix alone never certifies acceptance. If  $U' = 0$ , the answer is  $G = 1$ . Factor only this dense  $G$  and compute each returned factor's exact multiplicity by the preceding procedure. This is the deterministic univariate specialization of classical reconstruction [1].

## 2.3 Characteristic- $p$ descent without sparse division

Strip  $X^a$ , where  $a = \min n_i$ , and record its multiplicity separately. Maintain a polynomial  $F = U/V$  with  $U$  sparse,  $V$  dense,  $V \mid U$ , and  $U(0)V(0) \neq 0$ . Initially  $U = f/X^a$  and  $V = 1$ . Use the visible factors of  $U$  and dense factorization of  $V$  to form

$$A_U = \prod_g g^{e_U(g) \bmod p}, \quad A_V = \prod_g g^{e_V(g) \bmod p}.$$

Here  $e_U, e_V$  are exact multiplicities. There are unique polynomials, including scalar roots, such that  $U = A_U H_U^p$ ,  $V = A_V H_V^p$ . The dense polynomial  $H_V$  is found by exact division and coefficient inverse Frobenius. Since  $e_U(g) \geq e_V(g)$ ,

$$e_H(g) = \lfloor e_U(g)/p \rfloor - \lfloor e_V(g)/p \rfloor \geq 0, \quad H = H_U/H_V \in K[X].$$

Its radical is contained in that of  $F$ . This does not assert that each multiplicity of  $F$  is simply floor-divided: carries may occur.

Define the constant section  $C_0(\sum c_n X^n) = \sum c_{pn} Y^n$  and let  $\sigma$  be coefficient Frobenius, leaving the indeterminate fixed. The classical pull-out identity [5] gives

$$C_0(U) = C_0(A_U)\sigma(H_U).$$

Because  $A_U(0) \neq 0$ , set

$$U_{\text{new}} = \sigma^{-1}C_0(U), \quad W = \sigma^{-1}C_0(A_U), \quad V_{\text{new}} = WH_V.$$

Then  $H = U_{\text{new}}/V_{\text{new}}$  is a polynomial and  $V_{\text{new}} \mid U_{\text{new}}$ . Both constant terms remain nonzero. The new numerator has at most  $t$  terms and degree at most  $\lfloor \deg U/p \rfloor$ . Inverse Frobenius must act on *both* sections.

Let  $D_F$  be the radical degree of  $F$  and  $v = \deg V$ . The radical degree  $D_U$  of  $U = FV$  is at most  $D_F + v$ . Consequently

$$\begin{aligned} \deg V_{\text{new}} &\leq (\deg A_U + \deg V)/p \\ &\leq ((p-1)(D_F + v) + v)/p = v + (1 - 1/p)D_F \leq v + D_F. \end{aligned} \tag{3}$$

Also (1) gives  $\deg A_U \leq \min(p-1, t-1)D_U$ . Thus forming  $A_U$  densely has polynomial cost in  $t, D_U$ , even when  $p$  is binary-large. Radical containment implies  $D_F \leq D$  throughout, so after  $k$  descents  $\deg V \leq kD$ . Numerator degree, rather than degree of  $F$ , proves termination after at most  $b$  descents. A constant numerator forces a constant denominator.

The identity  $F = (A_U/A_V)H^p$  gives

$$e_F(g) = (e_U(g) \bmod p - e_V(g) \bmod p) + pe_H(g).$$

At level  $k$  add  $p^k$  times the signed residue difference to the factor ledger. Telescoping yields the exact nonnegative final multiplicities. Intermediate factors can be absent from the original input and must cancel. For example, in  $\mathbf{F}_2$ ,  $U = X^3 + X^2 + 1$  gives  $W = X + 1$  despite  $X + 1 \nmid U$ . Signed bookkeeping removes it at the following level. Use the original leading coefficient for the final unit.

## 2.4 Pseudocode and bit accounting

The preceding procedures give the following complete deterministic algorithm:

1. Normalize the sparse list, reject zero, strip  $X^a$ , set  $V = 1$ , the integer weight  $w = 1$ , and an empty signed factor ledger.
2. While the numerator is nonconstant, recover its visible denominator by (2), factor it densely, and find exact multiplicities. Factor  $V$  densely. Construct  $A_U, A_V, H_V$  as above.
3. Add  $w(e_U \bmod p - e_V \bmod p)$  for each occurring factor. Replace the numerator and denominator by

$$U \leftarrow \sigma^{-1}C_0(U), \quad V \leftarrow \sigma^{-1}C_0(A_U)H_V,$$

and replace  $w$  by  $pw$ . Continue step 2.

4. Combine equal factors, delete zero entries, restore  $(X, a)$  and the original leading coefficient. Check original-input valuations and the weighted degree if an independent certificate is desired.

At level  $k < b$ , the two dense calls have degrees at most  $(k+1)D$  and  $kD$ . Their total degree over all levels is  $O(b^2D)$ . Padé matrices have dimensions at most  $2 \max(1, (b+1)D)$  by that number plus one; a first known-factor quotient has dimension  $\delta \leq (b+1)D$  over  $K$ . Each coordinate always has  $L$  bits. The largest residue product degree is at most  $(t-1)(b+1)D$ . Multiplicities have  $O(b)$  bits; signed weights, including the final unused weight, have  $O(b + L + \log(b+1))$  bits; coefficient inverse Frobenius uses exponent  $p^{m-1}$  with  $O(mL)$  bits. Schoolbook polynomial arithmetic, elimination,

binary powering and exponent sorting are all polynomial in these displayed dimensions. With  $M = 1 + t + b + D + m + L$ , a deliberately loose nonoracle bound  $O(M^{24})$  bits covers them and the represented-field reductions below. Their prime calls contribute at most  $2b(b+1)D P((b+1)D, L)$ . This proves the theorem without a loop of numeric length  $p$  or  $N$ .

An independently checkable output certificate verifies distinct monic irreducibles, their exact multiplicities in the original sparse input,  $\sum e_i \deg g_i = N$ , and its leading coefficient. Divisibility and this degree equality prove the claimed factorization. It must not expand  $g_i^{e_i}$  when  $e_i$  is binary-large.

### 3 Explicit reduction for complete factorization

Let  $P(u, L)$  be a nondecreasing bit-cost bound for a deterministic complete factorization algorithm PF over  $\mathbf{F}_p$  on degree at most  $u$ .

**Theorem 2** (Established reduction with explicit oracle accounting). *There is a uniform deterministic reduction for complete factorization in the above representation with polynomial nonoracle overhead in  $n, m, L$ . The trace version uses at most  $mn$  calls to PF, each of degree at most  $n$ , with sum of queried degrees at most  $mn^2$ . An alternative uses at most  $n$  calls and sum of degrees at most  $n^2$ . Neither version enumerates the field or uses an integer-factorization or primitive-root oracle.*

#### 3.1 Fixed algebras and trace separation

First suppose  $f$  is monic squarefree of degree  $u$ . If it has  $s$  irreducible components, the Chinese remainder theorem gives

$$A = K[X]/(f) \simeq \prod_{a=1}^s \mathbf{F}_{p^{mu_a}}, \quad \sum_a u_a = u, \quad s \leq u.$$

With  $q = p^m$ ,  $q$ -Frobenius is  $K$ -linear. Its fixed algebra  $B_q = \ker(\text{Frob}_q - I)$  is  $K^s$ . Form its  $u \times u$  matrix and compute a kernel basis  $b_1, \dots, b_s$  by exact elimination. For  $0 \leq j < m$  form

$$c_{ij} = \sum_{k=0}^{m-1} (t^j b_i)^{p^k} \in A.$$

Each component of  $b_i$  belongs to  $K$ , so each component of  $c_{ij}$  is  $\text{Tr}_{K/\mathbf{F}_p}(t^j b_i(a)) \in \mathbf{F}_p$ . This is a quotient-algebra power computation, not a coefficientwise trace on arbitrary elements of  $A$ .

The trace polynomial  $\sum_{k=0}^{m-1} Y^{p^k}$  is nonzero and has degree  $p^{m-1} < p^m$ , so trace is nonzero, including when  $p \mid m$ . For every  $\delta \neq 0$ , multiplication by  $\delta$  is bijective; therefore  $y \mapsto \text{Tr}(\delta y)$  is nonzero. Some power-basis coordinate  $y = t^j$  detects it. Two different coordinate maps on  $B_q = K^s$  differ on some  $b_i$ ; hence some  $c_{ij}$  separates every pair of components. Using just  $\text{Tr}(b_i)$  would not justify this claim:  $\text{Tr}(1)$  can vanish.

If  $c$  has component values  $a_1, \dots, a_s \in \mathbf{F}_p$ , its monic minimal polynomial, over either  $K$  or  $\mathbf{F}_p$ , is exactly

$$\mu_c(Z) = \prod_{a \in \{a_1, \dots, a_s\}} (Z - a).$$

A polynomial annihilates  $c$  precisely when it vanishes at each coordinate; this proves the formula, squarefreeness and degree at most  $s$ . Find  $\mu_c$  by the first dependence among  $1, c, \dots, c^s$  in the  $u$ -dimensional  $K$ -space  $A$ . Its coefficients lie in the canonical  $\mathbf{F}_p$  and can be converted after checking

the other basis coordinates vanish. Factor  $\mu_c$  with PF. The returned factors are distinct linear polynomials and give its occurring values. Refine each current factor block  $g$  by  $\gcd(g, c - a)$  for those values  $a$ . These gcds partition its component support. After all  $ms$  coordinates no block contains two components, so every block is irreducible.

### 3.2 Independent prime-Frobenius variant

The  $p$ -power map is  $\mathbf{F}_p$ -linear, generally not  $K$ -linear. On the  $um$ -dimensional space with basis  $t^j X^i$ , form  $\text{Frob}_p - I$  as an  $um \times um$  matrix over  $\mathbf{F}_p$ . Its kernel  $B_p$  is  $\mathbf{F}_p^s$ , because the  $p$ -fixed elements of every finite field component are exactly  $\mathbf{F}_p$ . A basis of  $B_p$  separates all coordinate pairs. For each of its  $s$  basis vectors compute the first power relation over  $\mathbf{F}_p$  and apply the same minimal-polynomial factorization and gcd refinements. This proves correctness independently of the trace pairing and uses only  $s$  prime calls.

For one squarefree input, the trace version uses at most  $ms^3$  gcds and the direct version at most  $s^3$ . A first-dependence system has  $u$  rows over  $K$ , or  $um$  rows over  $\mathbf{F}_p$ , and at most  $s + 1$  columns. A loose count for the trace version is  $O(m^2 u^5 (L + 1))$  operations in  $K$ . For the direct variant one may bound the cost by

$$O(mu^3 L + u^5) \text{ operations in } K, \quad O(m^3 u^3 + mu^4) \text{ operations in } \mathbf{F}_p.$$

These counts include powering, exact elimination, dependence tracking, root-product checks and gcds. The prime calls add at most  $msP(s, L)$  or  $sP(s, L)$  bits respectively. Coefficients remain canonical field coordinates; no coefficient-height growth occurs in elimination. Schoolbook arithmetic and polynomial extended Euclid bound a  $K$  operation by  $O(m^2(L + 1)^3)$  bits, so these are uniform bit-polynomial reductions. The reference program's simple repeated dependence solves fit within the looser polynomial bounds; operation counters are not a runtime guarantee.

### 3.3 Multiplicity and inseparability

Normalize  $f$  by its leading coefficient. The standard squarefree procedure sets  $C = \gcd(f, f')$ ,  $W = f/C$ , and  $i = 1$ . While  $W \neq 1$ , set  $Y = \gcd(W, C)$ , emit  $(W/Y, i)$  if nonconstant, then set  $W = Y$ ,  $C = C/Y$ , and increment  $i$ . If  $C \neq 1$  after this loop, its remaining multiplicities are divisible by  $p$ ; its derivative is zero. For  $C = \sum c_k X^k$  its unique  $p$ th root is

$$H = \sum_j c_{pj}^{p^{m-1}} X^j.$$

Indeed  $a^{p^m} = a$  in  $K$ . Recurse on  $H$  and multiply the returned multiplicities by  $p$ . Copying coefficients works only when  $m = 1$ .

To check this procedure, fix a factor of multiplicity  $e$ . If  $p \nmid e$ , it initially occurs once in  $W$  and  $e - 1$  times in  $C$ , and is emitted at step  $i = e$ . If  $p \mid e$ , it remains wholly in  $C$  and passes to recursion with multiplicity  $e/p$ . This proves the result by induction. The emitted squarefree products have disjoint irreducible supports and sum of degrees at most  $n$ . Recursive input degrees sum to at most  $2n$ . All gcd loops have polynomial cost; coefficient-root exponents have  $O(mL)$  bits, not  $p^{m-1}$  iterations. Summing the squarefree call bounds gives exactly the bounds in the theorem. The argument covers  $p = 2$ ,  $m = 1$  and wholly inseparable inputs.

For checking  $h$ , compute successive  $T^{p^j} \bmod h$ , check the gcd with  $T^{p^j} - T$  for every  $j \leq \lfloor m/2 \rfloor$ , and check  $T^{p^m} = T \bmod h$ . A reducible degree- $m$  polynomial has an irreducible factor of degree at most  $m/2$ , so the gcds detect it, also with repeated factors. This uses  $m$  binary modular powers and no integer factorization.

## 4 Prescribed-degree construction

Shoup [10, Theorem 3.1] gives a deterministic reduction from constructing an irreducible degree- $N$  polynomial over  $\mathbf{F}_p$  to PF, with polynomial overhead in  $N, L$ . Theorem 4.1 gives the analogous arbitrary-base implication. For clarity an alternative known lift is as follows. Construct  $H \in \mathbf{F}_p[X]$  irreducible of degree  $N = md$ , factor it over  $K$  by the preceding theorem, and take the lexicographically first factor. The  $p$ -Frobenius orbit of a root has size  $N$ .  $p^m$ -Frobenius acts by adding  $m$  modulo  $N$ , so every orbit has size  $N/\gcd(N, m) = d$ . Every factor therefore has degree  $d$  and multiplicity one. This proves the lift without finding an isomorphism or primitive root. It is the same lift as Rai’s Algorithm 2 [11]. For  $d = 1$  return  $X$ ; for  $m = 1$  direct prime construction suffices. The extension is the explicit quotient  $K[X]/(g)$  for the output  $g$ .

The supplementary construction audit makes Shoup’s oracle use explicit: trial division factors the numeric output degree  $N$ ; each prime-power construction has prime-field algebra dimension at most  $N^2$ ; oracle polynomial degrees are at most  $N^2$ , and there are  $O(NL \log(N + 1))$  calls. The coefficient slots always have  $L$  bits. The bound includes  $p$ -power Artin–Schreier towers, odd prime-power binomials and the special 2-power cases. It never factors  $p - 1$ . Trial division of  $N$  is permitted by dense output size and is not an integer-factorization oracle on arbitrary binary inputs. The 2-power exception cannot be omitted: over  $\mathbf{F}_3$ ,

$$X^4 + 1 = (X^2 + X + 2)(X^2 + 2X + 2),$$

although  $-1$  is nonsquare. The supplied construction uses the quadratic extension in that case, following the established binomial criterion. Factoring  $H$  over  $K$  has input size  $O(dm^2L)$ , polynomial in the original field and dense degree- $d$  output sizes. Thus the whole construction is bit-polynomial whenever  $P$  is polynomial.

## 5 Upstream input and resulting scope

**Corollary 3** (Consequence of OpenAI’s prime-field theorem). *Using [12, Theorem 1.1], complete factorization over the represented  $K$ , on either dense input or sparse input with the dense factor output specified above, and degree- $d$  irreducible construction are uniform deterministic algorithms with bit complexity polynomial in their respective input/output sizes. They require no GRH, randomness, integer-factorization oracle or primitive-root oracle.*

*Proof.* Substitute the cited prime-field bound  $P(u, L) = O(((u+1)L)^{10^{12}})$  into the established dense reductions and Theorem 1. Every oracle degree, number of calls and nonoracle representation is bounded by a fixed polynomial in the corresponding dense or sparse input/output parameters stated above.  $\square$

This is a consequence of the cited proof, not an independent proof of its analytic or geometric breakthrough. Family 142’s Theorem 1.2 is an auxiliary-prime reduction with numeric prime-size parameter  $E$ . Its unconditional Theorem 1.1 imports family 029’s Theorem 1.2 [13]: all finite-order Hecke  $L$ -functions over every cyclotomic field containing  $\mu_{12}$  are zero-free in  $\Re s > 1 - 10^{-6}$ , with the principal pole permitted. The applications use  $F = \mathbf{Q}(\mu_{12q})$  and the cyclic extension  $F(p^{1/q})/F$ . Abelian zeta factorization and the smoothed explicit formula produce auxiliary primes  $\ell \leq cB^{20000000}$ , where  $B = 20 + (n + 1)(L + 1)$ . A Dirichlet-only theorem does not suffice.

We inspected the actual manuscripts at the pinned repository commit

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The audit covers source hashes, the required theta, reflection, Poisson and Mellin arguments, and the algebraic norm, divisor and splitting proof. Separate automated adversarial audits found no substantive gap in those inspected dependencies. Limited finite checks of auxiliary tables, local Gauss/Euler identities and exact exponent margins are supplied; they do not prove an analytic zero-free theorem. The huge prime-field branch was not executed. Family 003’s actual Lean scope is Dirichlet and Eisenstein-field Hecke nonvanishing; it does not cover the needed varying cyclotomic input. No relevant Lean build was reproduced here and this follow-on is not claimed formally verified.

For optional fixed- $r$  extraction, factor the dense polynomial  $X^r - a$  and return its linear-factor roots. Complexity is polynomial in numeric  $r, m, L$ , hence in  $m, L$  for each fixed  $r$ . No polynomial-in- $\log r$  claim follows. This is merely a factoring consequence; quadratic nonresidue and square-root consequences already in the release are not relabeled as new results.

## 6 Reproduction, limitations and disclosure

The supplement contains exact reduction pseudocode, source/dependency ledgers, proof-node audits and standard-library Python implementations. The bundled prime oracle enumerates  $\mathbf{F}_p$  only in bounded small examples; its  $O(p)$  cost is not polynomial in  $\log p$  and cannot substantiate prime-field derandomization. Independent trial-division comparison checks 377 small monic inputs; a second implementation cross-checks 346 cases. Examples cover characteristic two, nontrivial inverse coefficient Frobenius, repeated factors,  $m = 1$ , constants and rejected zero/oracle errors. The manuscript makes no assertion of practical efficiency or first priority. The sparse implementation additionally compares 960 small monic inputs with dense factorization and checks huge binary exponents, signed cancellations and nontrivial extension coefficients. Separate independent prototypes check the multiplicity and full-descent reductions. These are finite checks, not universal complexity tests. The supplemental priority audit separates the inherited individual mechanisms from the sparse/dense degree invariant and compares exact primary statements. The earlier dense-only research record was withheld from Zenodo as inherited; the present theorem adds the all-multiplicity sparse output bound. No claim about general arithmetic circuits, sparse-only output size or polynomial cost in binary prescribed construction degree is made.

AI tools were used extensively in research, drafting and verification. Automated adversarial reviews are not conventional human peer review or refereeing; this work has not undergone such review. Alec Kriebel is the sole listed author; no affiliation or outside endorsement is asserted. Upstream AI-produced mathematical work is attributed to OpenAI as supplied by its manuscript-specific citation information.

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